

Luke McCartin, PhD

Postdoc, Smithsonian National Museum of Natural History, Washington, D.C., USA

Department of Invertebrate Zoology

Email: McCartinL@SI.edu

Curriculum Vitae

Education

- 2019 – 2024 **PhD, Biological Sciences**, GPA: 4.00
Department of Biological Sciences, Lehigh University, Bethlehem, PA, USA
- 2013 – 2017 **B.S., Biological Sciences**, GPA: 3.68
Biology Department, University of Vermont (UVM), Burlington, VT, USA

Research Appointments

- December 2025 – present **Postdoctoral Researcher**, Smithsonian National Museum of Natural History, Dept. of Invertebrate Zoology
- August 2024 – December 2025 **Post-doctoral Research Associate**, Lehigh University, Dept. of Biological Sciences
- January – May 2024 **Thorne Fellow**, Lehigh University, Dept. of Biological Sciences
- August – December 2023 **Smithsonian Pre-doctoral Research Fellow**, Smithsonian National Museum of Natural History, Dept. of Invertebrate Zoology
- June 2022 – August 2024 **PhD Candidate**, Lehigh University, Dept. of Biological Sciences
- August 2019 – June 2022 **Graduate Student**, Lehigh University, Dept. of Biological Sciences
- June 2017 – August 2019 **Research Assistant**, Woods Hole Oceanographic Institution Dept. of Biology – June 2017 to August 2019
- Summer 2016 **Summer Undergraduate Research Fellow**, University of Vermont Dept. of Biology
- May 2014 – August 2015 **Biological Science Technician**, U.S. Department of Agriculture Animal and Plant Health Inspection Service

Publications (18)

In review

1. Herrera, S., Govindarajan, A.F., Andruszkiewicz Allan, E., Francolini, R., Frates, E., **McCartin, L.**, Pittoors, N.C., Sengthep, M., Stover, S., Vohsen, S. and Yang, N. Depth-dependent eDNA abundances across ecosystems inform deep-sea sampling strategies. *bioRxiv*, pp.2026-05. <https://doi.org/10.64898/2026.05.12.724363>
2. Pittoors, N.C., Tweedt, S.M., **McCartin, L.J.**, Vohsen, S.A., Lopera, L., Mihalek, S., Lai, J., Durkin, K., Weigt, L., Nuttall, M.F. and Bracco, A. Environmental filtering shapes patch dynamics across isolated mesophotic reefs. *bioRxiv*, pp.2025-11. <https://doi.org/10.1101/2025.11.02.686126>

Published

3. Horowitz, J., Opresko, D.M., Morrissey, D., Polit, V., Shull, H.C., Cerqueira, T., **McCartin, L.J.**, Vohsen, S.A., Carreiro-Silva, M. and Quattrini, A.M. (2026). The ghost of glaberrima: descriptions of two new *Leiopathes* species. *Coral Reefs*, pp.1-23. <https://doi.org/10.1007/s00338-026-02903-1>
4. **McCartin, L.J.**, Vohsen, S.A., Wood, A.L., Horowitz, J., Orozco-Juarbe, J.J., Pittoors, N., Morrissey, D., Vaga, C.F., Hansel, C.M., Collins, A.G. and Quattrini, A.M. (2026). Accounting for Intra-and Intergenomic Sequence Variation in Reference Barcodes Improves eDNA Metabarcoding Biodiversity Assessment. *Molecular Ecology Resources*, 26(3), p.e70130. <https://doi.org/10.1111/1755-0998.70130>
5. **McCartin, L.J.**, Govindarajan, A.F., McDermott, J.M. and Herrera, S. (2025). Limited Dispersal of Benthic Environmental DNA From a Subtropical Mesophotic Shelf-Edge Bank. *Environmental DNA*, 7(5), p.e70204. <https://doi.org/10.1002/edn3.70204>
6. Vaga, C.F., Cairns, S.D., **McCartin, L.J.**, Herrera, S., Kitahara, M. & Quattrini, A. M. (2025). Phylogenetic position of the reef-builder *Madrepora carolina* and the genus *Thalamophyllia* (Anthozoa, Scleractinia): new additions for the family Agariciidae. *Marine Biodiversity*. 55, 69. <https://doi.org/10.1007/s12526-025-01552-x>
7. Wood, A., **McCartin, L.**, Herrera, S., Quattrini, A. M., & Collins, A. G. (2025). Anthozoan eDNA Primer Set Characterizes Ctenophora and Medusozoa Inhabiting Mesophotic and Deep Waters off the Gulf Coast of the Southern USA. *Frontiers in Marine Science*, 12, 1535149. <https://doi.org/10.3389/fmars.2025.1535149>
8. Quattrini, A. M., Morrissey, D. & **McCartin, L. J.** (2025) A new soft coral species from the Gulf of Mexico (Octocorallia: Scleractyonacea: Parasphaerascleridae). *Zootaxa* 5601.3 (2025): 545-557. <https://doi.org/10.11646/zootaxa.5601.3.8>
9. Horowitz, J., Barajas, M., **McCartin, L. J.**, Vohsen, S. A., & Herrera, S. (2025) Description of a new species of *Stauropathes* (Anthozoa, Antipatharia, Schizopathidae) from Puerto Rico. *ZooKeys* 1231 (2025): 331-346. <https://doi.org/10.3897/zookeys.1231.136967>
10. **L. J. McCartin**, E. Saso, S. A. Vohsen, N. C. Pittoors, P. Demetriades, C. S. McFadden, A. M. Quattrini & S. Herrera. (2024). Nuclear eDNA Metabarcoding Primers for Anthozoan Coral Biodiversity Assessment. *PeerJ* 12: e18607. <https://doi.org/10.7717/peerj.18607>
11. Quattrini, A. M., **McCartin, L. J.**, Easton, E. E., Horowitz, J., Wirshing, H. H., Bowers, H., Mitchell, K., Sei, M., McFadden, C. S., & Herrera, S. (2024). Skimming genomes for systematics and DNA barcodes of corals. *Ecology and Evolution* 14(5), e11254. <https://doi.org/10.1002/ece3.11254>
12. T. P. Heyl, G. Panieri, D. J. Fornari, R. Mattingsdal, S. Sauer, H. Yao, **L. McCartin**, E. McElwee, & T.M. Shank. (2023). Implications of transient methane flux on associated biological communities in high-arctic seep habitats, Storbanken, Norwegian Barents Sea. *Deep Sea Research Part I: Oceanographic Research Papers* 201: 104156. <https://doi.org/10.1016/j.dsr.2023.104156>
13. S. Herrera, W. W. Chadwick Jr., M. G. Jackson, J. Konter, **L. McCartin**, N. Pittoors, E. Bushta, & S. G. Merle. (2023). From basalt to biosphere: Early non-vent community succession on the erupting Vailulu'u deep seamount. *Frontiers in Marine Science* 10 <https://doi.org/10.3389/fmars.2023.1110062>
14. A. F. Govindarajan, A. Adams, E. Allan, S. Herrera, A. Lavery, J. Llopiz, **L. McCartin**, D. R. Yoerger, & W. Zhang. (2023). Advances in Environmental DNA Technology for Observing Ocean Twilight Zone Animal Biodiversity. *2nd Annual Supplement to Oceanography: Frontiers in Ocean Observing* <https://doi.org/10.5670/oceanog.2023.s1.27>
15. A. F. Govindarajan, **L. J. McCartin**, A. Adams, E. Allan, A. Belani, R. Francolini, J. Fujii, D. Gomez-Ibañez, A. Kukulya, F. Marin, K. Tradd, D. R. Yoerger, J. M. McDermott, S. Herrera. (2022). Improved biodiversity detection using a large-volume environmental DNA sampler with in situ filtration and

implications for marine eDNA sampling strategies. *Deep-sea Research Part I – Topical Studies in Oceanography* 189: 103871. <https://doi.org/10.1016/j.dsr.2022.103871>

16. **L. J. McCartin**, S. Vohsen, S. Ambrose, M. Layden, C. McFadden, E. Cordes, J. McDermott, S. Herrera. (2022). Temperature Controls eDNA Persistence Across Physicochemical Conditions in Seawater. *Environmental Science & Technology* 56, 12, 8629–8639. <https://doi.org/10.1021/acs.est.2c01672>
17. **L. J. McCartin**, N. A. Nasser, & A. K. Brody. (2019). Potential Consequences of Hosting an Ant-tended Treehopper, *Publilia concava*, for Tall Goldenrod, *Solidago altissima*. *American Journal of Undergraduate Research* 15, 4. <https://doi.org/10.33697/ajur.2019.002>
18. J. R. Gould, B. Aflague, T. C. Murphy, **L. McCartin**, J. S. Elkington, K. Rim, & J. J. Duan. (2018). Collecting Nontarget Wood-Boring Insects for Host-Specificity Testing of Natural Enemies of Cerambycids: A Case Study of *Destarcus helophoroides* (Coleoptera: Bothrideridae), a Parasitoid of the Asian Longhorned Beetle (Coleoptera: Cerambycidae). *Environmental Entomology* 47, 6, 1440-1450. <https://doi.org/10.1093/ee/nvy121>

Video Data Product

1. Herrera, S. & McCartin, L. (2024). Mesophotic and Deep Benthic Communities: Annotations and images of corals and sponges from the northern Gulf of Mexico from 2017-07-20 to 2021-08-15 (NCEI Accession 0299813). NOAA National Centers for Environmental Information. <https://www.ncei.noaa.gov/archive/accession/0299813>

Awarded Fellowships

- | | |
|------|---|
| 2024 | Lehigh Department of Biological Sciences – Thorne Fellowship, \$13,077 USD |
| 2023 | Smithsonian Instruction Fellowship Program – Predoctoral Fellowship, \$20,104 USD |
| 2016 | UVM Office of Undergraduate Research – Summer Undergraduate Research Fellowship: Summer Research Award, \$4,879 USD |

Competitive Travel Awards/Small Grants

- | | |
|------|---|
| 2025 | COBRA Early Career Accelerator Stipend – Training award for presentation/attendance at the 17 th Deep-Sea Biology Symposium, \$1,486 USD |
| | Deep-Sea Biology Society – Travel award for presentation at the 17 th Deep-Sea Biology Symposium, \$971 USD |
| 2019 | Deep-Sea Biology Society – Travel award for presentation at the 7 th International Symposium on Deep-sea Corals, \$927 USD |

Oceanographic Research Expeditions

- | | |
|------|--|
| 2025 | <u>MDBC PS26-09 ARMS Recovery</u> , R/V Point Sur, Gulfport, Mississippi |
| 2024 | <u>E Mamana Ou Gataifale I</u> , E/V Nautilus, Pago Pago, American Samoa |
| 2023 | <u>Health Diagnostics of Deep-sea Corals</u> , R/V Falkor (too), San Juan, Puerto Rico |
| | AT50-07, R/V Atlantis, Puntarenas, Costa Rica |
| 2022 | SALT3, R/V Point Sur, Gulfport, Mississippi |
| | <u>Illuminating Biodiversity in Deep Waters of Puerto Rico</u> , NOAA Ship Nancy Foster, San Juan, Puerto Rico |

- 2021 [Connectivity of Coral Ecosystems](#), *R/V Point Sur*, Gulfport, Mississippi
- 2019 **Environmental DNA Exploration**, *R/V Manta*, Galveston, Texas
- 2017 [Deep-sea Corals of PIPA](#), *R/V Falkor*, Phoenix Islands, Kiribati

Professional Development

- 2025 Teacher Development Series Parts I & II, Center for Innovation in Teaching and Learning, Lehigh University.
- 2024 Workshop on Molecular Evolution, University of Chicago's Marine Biological Laboratory, Woods Hole, MA.
- Deep Submergence Science Committee (DeSSC) New Users Program, New Orleans, LA.
- 2022 Strategies and Techniques for Analyzing Microbial Population Structures (STAMPS), University of Chicago's Marine Biological Laboratory, Woods Hole, MA.

Invited Scientific Talks

- 2026 DOOS 2026 Annual Meeting, Determining the Needs for Deep Biomolecular Observing, *Comprehensive Reference Barcode Databases Improve Deep-sea eDNA Biodiversity Assessment*
- 2024 Johns Hopkins Applied Physics Laboratory, 3rd National Workshop on Marine eDNA, *Temperature Controls the Degradation Rate of Marine eDNA*
- Smithsonian National Museum of Natural History Department of Invertebrate Zoology, "No Bones" Seminar Series, *Pairing Environmental DNA Sequencing and Genome Skimming to Characterize Anthozoan Coral Biodiversity in the Deep Western Atlantic*
- 2022 Smithsonian National Museum of Natural History Department of Invertebrate Zoology, "No Bones" Seminar Series, *Environmental DNA Sequencing to Study Deep Coral Communities*

Conference Presentations

- 2025 **Talk:** Luke McCartin, Nicole Pittoors, Penny Demetriades, Samuel Vohsen, Sarah Tweedt, Chris Meyer, Santiago Herrera. *Environmental DNA (eDNA) Distinguishes Benthic Communities in the northwestern Gulf of Mexico Mesophotic Reefs and Banks*. Benthic Ecology Meeting. Mobile, Alabama.
- Talk:** Luke McCartin, Annemarie Wood, Samuel Vohsen, Jeremy Horowitz, Allen Collins, Colleen Hansel, Andrea Quattrini & Santiago Herrera. *Characterizing the Biodiversity and Bathymetric Distributions of Caribbean Deep-Sea Corals through eDNA*. 17th Deep-Sea Biology Symposium. Hong Kong SAR, China.
- 2024 **Talk:** Luke McCartin, Emma Saso, Sam Vohsen, Nicole Pittoors, Penny Demetriades, Catherine McFadden, Andrea Quattrini & Santiago Herrera. *New eDNA Meta-barcoding Primers to Elucidate the Distribution of Diverse Anthozoan Corals*. Ocean Sciences Meeting. New Orleans, Louisiana.
- 2023 **Talk:** Luke McCartin, Emma Saso, Sam Vohsen, Nicole Pittoors, Penny Demetriades, Catherine McFadden, Andrea Quattrini & Santiago Herrera. *Nuclear*

Environmental DNA Sequencing to Study Mesophotic and Deep Anthozoan Corals.
8th International Symposium on Deep-sea Corals. Edinburgh, Scotland.

Talk: Luke McCartin, Emma Saso, Sam Vohsen, Nicole Pittoors, Penny Demetriades, Catherine McFadden, Andrea Quattrini & Santiago Herrera. *Nuclear Environmental DNA Sequencing to Study Mesophotic and Deep Anthozoan Corals.* NOAA Open Ocean Restoration Plan, Natural Resource Damage Assessment. Mesophotic and Deep Benthic Communities. 3rd Habitat Assessment and Evaluation Workshop. Smithsonian NMNH, Washington, DC.

2022 **Talk:** L. McCartin, A. F. Govindarajan, J. McDermott, S. Herrera. *Zooplankton eDNA community composition is structured by depth in the epipelagic Gulf of Mexico.* SCOR MetaZooGene Symposium. Dublin, Ireland.

Poster: L. J. McCartin, S. Vohsen, S. Ambrose, M. Layden, C. McFadden, E. Cordes, A. F. Govindarajan, J. McDermott, S. Herrera. *Temperature Controls Environmental DNA Persistence Across Physicochemical Conditions in Seawater.* 2nd National Workshop on Marine Environmental DNA. Costa Mesa, CA.

Talk: L. McCartin, S. Ambrose, S. Vohsen, E. Saso, E. Cordes, A. Govindarajan, C. McFadden, A. Quattrini, D. Yoerger, J. McDermott & S. Herrera. *Environmental DNA Sequencing to Study Deep Coral Reefs in the Gulf of Mexico.* The Gulf of Mexico Conference. Baton Rouge, LA.

2021 **Talk:** L. McCartin, S. Vohsen, S. Ambrose, A. Hallaj, M. Layden, E. Cordes, C. McFadden, J. McDermott & S. Herrera. *Physicochemical Controls on the Persistence of Marine Environmental DNA.* ASLO 2021 Virtual Aquatic Sciences Meeting.

2019 **Lightning talk & poster:** L. McCartin, E. McElwee, T. Heyl, S. Auscavitch, E. Cordes, R. Rotjan & T. M. Shank. *Patterns of Coral Symbiont Distributions Among Deep Waters of Atolls, Islands, Reefs and Seamounts in the Phoenix Islands Protected Area.* 7th International Symposium on Deep-sea Corals. Cartagena, Colombia.

2017 **Poster:** L. McCartin, N. Nasserri & A. K. Brody. *The Ant-hemipteran Mutualism on Goldenrod.* University of Vermont Student Research Conference, Burlington, VT.

Academic Service

- **Co guest editor:** *Ocean Observing Supplement to Journal of Oceanography Volume 36 – Frontiers in Ocean Observing: Emerging Technologies for Understanding and Managing a Changing Ocean.* Topic 6: *Environmental DNA Technology.* <https://doi.org/10.5670/oceanog.2023.s1.1>
- **Manuscript reviewer:** *Environmental Science & Technology, Scientific Reports, Molecular Ecology, Molecular Ecology Resources, Marine Biology, Environmental DNA, Biodiversity & Conservation, Deep-sea Research Part II, Regional Studies, Environmental Biology of Fishes, Environmental Science & Pollution Research, Journal of Thermal Biology, Aquatic Conservation: Marine and Freshwater Ecosystems*
- **Ad hoc proposal reviewer:** NSF BIO-OCE program
- **Proposal reviewer:** Lehigh University College of Arts & Sciences Dean's Fellowship

Societal Memberships

- Deep-sea Biology Society
- Marine Technology Society

Mentorship

- Luis Caceres – Lehigh undergrad [RARE](#) and [STEM-SI](#) program student[§]
- Jacob Cardoza – Lehigh undergraduate student
- Eloisa Dadalt – Smithsonian predoctoral fellow
- Maia Clayton – Lehigh undergraduate work study student
- Penny Demetriades – *Lehigh undergraduate student*^{§†}
- Carly Hecht – Lehigh undergraduate student
- Samantha Kratman – Smithsonian intern
- Mary Mahoney – Lehigh undergrad STEM-SI program student[§]
- Petra Oganovich – Lehigh undergraduate student^{||¶}
- Milan Sengthep – *Lehigh PhD student*^{†‡}
- Annemarie Wood – *Smithsonian intern*^{*†‡}

* - first-authored publication

† - co-authored publication

‡ - presented work at national conference

§ - presented at university symposia

|| - presented at Lehigh undergraduate research symposiums.

¶ - received internal university grant funding totaling \$7,592.

Teaching Positions

Fall 2021	Lehigh Biology Graduate TA: Evolution
Spring 2021	Lehigh Biology Graduate TA: Biology Core III
Spring 2020	Lehigh Biology Graduate TA: Biostatistics/Microbiology Lab
Fall 2019	Lehigh Biology Graduate TA: Biology Core III
June 2018 – Aug. 2019	Math & Science Tutor: Solutions Learning Center
Spring 2017	Biology Tutor: UVM Center for Academic Success
Fall 2016 – Spring 2017	Undergraduate TA: UVM Introductory Biology Labs

Course Development

- 2024 3-part module: *Hands-on Tutorials for DNA Metabarcoding Analysis*. Molecular Ecology Course. Lehigh University.

Educational Seminars and Classroom Lectures

- 2026 Seminar: Intern Data Carpentry workshop. *Data visualization with ggplot2*. Smithsonian Environmental Research Center.
- Guest Lecture: Dr. Layla Al-Shaer's Ecology course. *Illuminating the Deep-sea with Environmental DNA*. Lafayette College.
- 2025 Guest Lecture: Dr. Santiago Herrera's Marine Biology course. *Illuminating Deep-sea Ecology with Environmental DNA*. Lehigh University.
- Seminar: *Environmental DNA and its power for studying biodiversity in the deep sea*. PEERSIDE mentorship program. Florida Institute of Oceanography.
- 2023 Seminar: *Coral Forensics in the Deep Flower Garden Banks*. Seaside Chats Webinar Series: National Marine Sanctuaries Education.
- 2021 Seminar: *Using Molecular Tools to Study Marine Biology*. Marine Biodiversity and Conservation course. Sea Semester. Woods Hole, MA.
- 2019 Seminar: *Bird Banding on Wing Island*. Cape Cod Museum of Natural History Field Guide Training Program. Brewster, MA.

Outreach

- 2026 **World Ocean's Day**, Ocean Hall, Smithsonian National Museum of Natural History (NMNH)
- Sanibel Board Game Night**, Ocean Hall, Smithsonian NMNH
- 2025 **NOAA Ship-to-shore engagement**, *R/V Point Sur*
- 2024 **OEI NOAA Science Seminar: Midwater Exploration through Multi-Vehicle Operations**, *E/V Nautilus*
- Expert Is in Takeover**, Ocean Hall, Smithsonian NMNH
- Blog post**, *New Methodology for eDNA Collection to Study Deep-sea Ecosystems of Puerto Rico*, on the NOAA OER *Illuminating Biodiversity* expedition site
- 2023 **Seaside Chats Webinar Series: Coral Forensics in the Deep Flower Garden Banks**, National Marine Sanctuaries Education
- 2022 **Expert Is in Takeover**, Ocean Hall, Smithsonian National Museum of Natural History
- Science Fair Judge**, Southern Lehigh High School, Center Valley, PA
- 2019 **Blog post**, *Deep-sea Fish of PIPA* on the Schmidt Ocean Institute *Deep-sea Corals of PIPA* expedition site